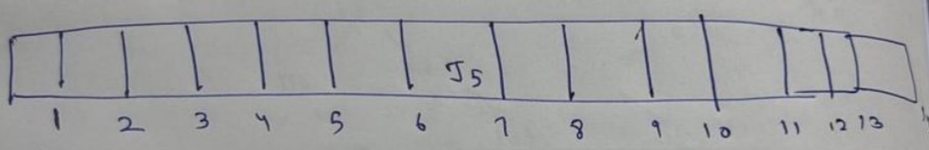
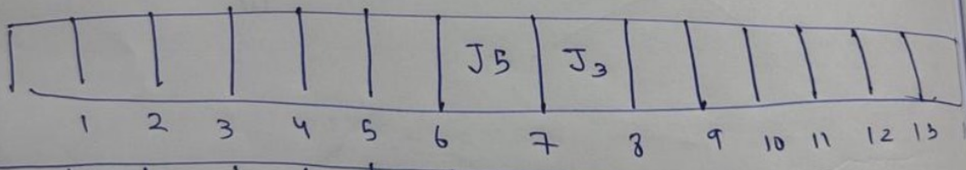
```c
#include <stdio.h>

struct Job {
    int id;
    int profit;
    int deadline;
};

int main() {
    int n, i, j;
    struct Job temp;

    printf("Enter number of jobs: ");
    scanf("%d", &n);

    struct Job jobs[n];

    // Input jobs
    for(i = 0; i < n; i++) {
        printf("Enter Profit and Deadline for Job %d: ", i + 1);
        jobs[i].id = i + 1;
        scanf("%d %d", &jobs[i].profit, &jobs[i].deadline);
    }

    // Sort jobs by profit (descending)
    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(jobs[i].profit < jobs[j].profit) {
                temp = jobs[i];
                jobs[i] = jobs[j];
                jobs[j] = temp;
            }
        }
    }

    int slot[n];
    int result[n];

    for(i = 0; i < n; i++)
        slot[i] = -1;
```

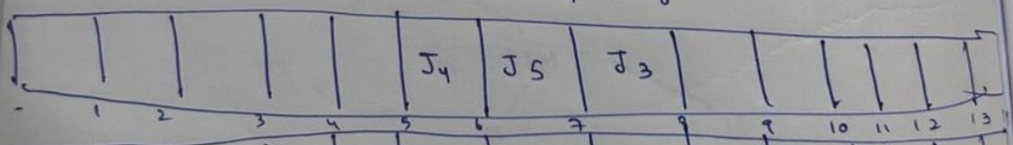
```
    // Job sequencing
    for(i = 0; i < n; i++) {
        for(j = jobs[i].deadline - 1; j >= 0; j--) {
            if(slot[j] == -1) {
                slot[j] = jobs[i].id;
                result[j] = jobs[i].profit;
                break;
            }
        }
    }

    int totalProfit = 0;

    printf("\nSelected Jobs:\n");
    for(i = 0; i < n; i++) {
        if(slot[i] != -1) {
            printf("Job %d ", slot[i]);
            totalProfit += result[i];
        }
    }

    printf("\nTotal Profit: %d\n", totalProfit);

    return 0;
}
```

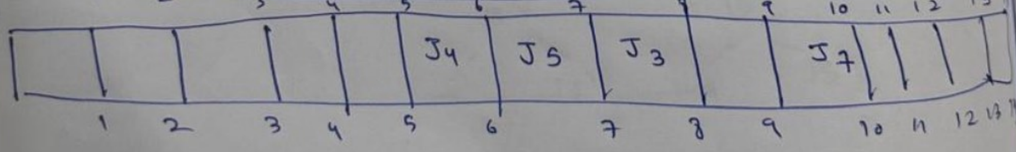
```
C:\Users\A JEEVITESWARA REDDY\OneDrive\Desktop>daa
Enter number of jobs: 14
Enter Profit and Deadline for Job 1: 22 3
Enter Profit and Deadline for Job 2: 19 3
Enter Profit and Deadline for Job 3: 29 8
Enter Profit and Deadline for Job 4: 28 6
Enter Profit and Deadline for Job 5: 30 7
Enter Profit and Deadline for Job 6: 21 5
Enter Profit and Deadline for Job 7: 27 10
Enter Profit and Deadline for Job 8: 25 4
Enter Profit and Deadline for Job 9: 24 6
Enter Profit and Deadline for Job 10: 26 12
Enter Profit and Deadline for Job 11: 14 18
Enter Profit and Deadline for Job 12: 27 2
Enter Profit and Deadline for Job 13: 19 14
Enter Profit and Deadline for Job 14: 11 1

Selected Jobs:
Job 6 Job 12 Job 1 Job 8 Job 9 Job 4 Job 5 Job 3 Job 7 Job 10 Job 11 Job 13
Total Profit: 292
```

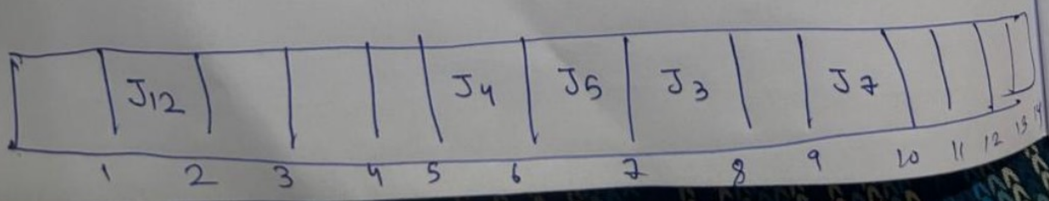
Job	Profit	Deadline
J_5	30	7
J_3	29	8
J_4	28	6
J_7	27	10
J_{12}	27	2
J_{10}	26	12
J_8	25	4
J_9	24	6
J_1	23	3
J_6	21	5
J_2	19	3
J_{13}	19	14
J_{11}	14	18
J_{14}	11	1

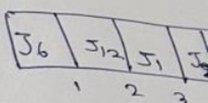












Job Profi

J_6	J_{12}	J_1	J_2	J_9	J_4	J_5	J_3		J_7		J_{10}	J_{11}	J_{13}
1	2	3	4	5	6	7	8	9	10	11	12	13	14

Job Profit → 292